

Grammar Without Reification: Grammar As Abstract Object

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Grammar and Reification

- Grammars are abstract. They are abstract objects.
- They are abstract because they rise above data, more than what we are exposed to.
- They are real, i.e. objects, because they enable us to do work.
- Minimally, that work is grammaticality.
- If something is grammatical, then it is meaningful, no matter how senseless that meaning is.

Maybe we can study both aspects at the same time.

- Grammars are abstract. They are abstract objects.
- That abstraction deserves to be studied without being reified to psychology, biology, logic, semiotics, etc.
- An abstract grammar reified is not abstract anymore.
- Further commitments down the road either exacerbates patch management in theory, or forces commitments to ontology external to language.
- However, linguistic capacity seems to be modular, at least representationally.
- **Linguistic categories** don't look or act like planning, action, image, dancing, or social categories.

- Mathematics studies abstraction, the kind and amount of abstraction to reach an **invariant**.
- We may reach **linguistic invariants** proceeding that way. A mathematical understanding of limits of linguistic variation.
- **Variants** are then consequences of invariants. That is linguistic diversity, the domain of typology.
- Can we do this without mathematicizing linguistics?
- Questions arise:
 - What does an abstract grammar look like?
 - What would it contain?
 - What would it do?

- Questions:
 - 1 What does an abstract grammar look like?
 - 2 What would it contain?
 - 3 What would it do?
- One proposal (Bozşahin 2025, studied in this course):
 - 1 Abstract grammars can be **monadic** (working with **function composition** only; its invariant as math of **connectedness**), consisting exclusively of linguistic elements with **categories**.
 - 1 Linguistic categories are **functions**, mathematically.
 - 2 Every category must contain two **command relations**, syntactic command and semantic command, to target grammaticality and consequent sense of meaningfulness. They are variants, i.e. consequences of the invariant.
 - 3 They target all **intralinguistic and crosslinguistic variation**.

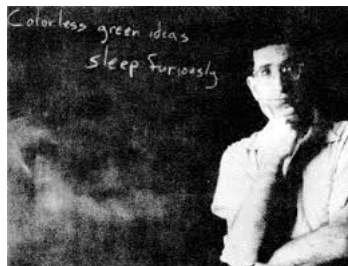
Potential niches for the variants

- **Gestaltists** (Wertheimer, 1924; Koffka, 1936; Köhler, 1938; Arnheim, 1954) were on to something, before disrupted by WWII.
- They were interested in wholistic understanding, not componential analyses. They were studying **values**.
- Recall Principle of Compositionality, often attributed to Frege (1879) (more on that later):
The meaning of the whole is determined by the meaning of the parts and the way they are combined.
- Gestaltists thought the other way around: The whole determines the parts that make it meaningful. The parts just fragment a complex expression (linguistic, visual, audio, etc.).
- As such, the whole is a function, manifestation of **connectedness** (i.e. composition). That's the idea behind the invariant.

What is Knowledge of Grammar?

- The answer depends on what we expect from grammar.
 - Minimally, we want it to decide grammaticality (i.e. syntactic configuration).
 - Maximally, we would want it to cover everything under the sun about knowledge of language.
- Is this a matter of personal choice?
 - Every child exposed to NL data in the critical period acquires a natural language.
 - There are systematic exclusions if she is not.
 - Exclusions from what? Where do we show that?
 - Every native speaker has a sense of meaningfulness of an expression which is consequent to it being grammatical.

Colorless green ideas sleep furiously.



- NL grammars are representational spaces. Is this space limited?
- What is a linguistic category? Label? abstract, concrete?
- Anderson (1976): Data from ergativity and mixed systems show that the category space cannot be universally captured morphologically or surface-categorially (e.g. case, N, V, A, P).
- Categories in a grammar can be abstract objects (see Katz 1985 for a bit of philosophy of linguistics).

Some grammars as conceived by typologists and field linguists:

Bunan (Tibeto-Burman)

Paiwan (Taiwan, Austronesian)

Three related concerns:

- 1 Can we make human grammars (their abstraction) determine grammaticality without external means?
- 2 Or do we need extra means? (minimal links, mapping theories for multi-structures, multiple independent computations, movement management, subjacency, synch conditions, reanalysis, recomputation etc.)
- 3 Our theories must be independently testable, not just by believers.

The concerns are critical: The more intermediaries there are, the more difficult it is to go from theory of grammar to models of grammar.

Scientific modellability is not a personal choice. That doesn't mean everyone has to model.

One consequent question:

Is it enough to determine grammaticality?
(syntactic well-formedness)

The answer will determine what goes in to any NL grammar.

What kind of meaningful categories must enter grammar?

Colorless green ideas sleep furiously.

- Wittgenstein
- Chomsky
- Leśniewski (1929)
- Ryle (1937)
- W: Language games can make it meaningful.
- C: It is already meaningful, because it is grammatical. What it lacks is sense.
- L: Categories are semantic in origin; no such thing as meaningless category.
- R: Categories are not for linguists and philosophers to second-guess the beholder. We try to understand how they do it.

- 1 If something is grammatical and senseful, we can think of world models for meaning.
 - Ungrammatical ones are not meaningful to begin with:
 - a. *Green sleep colorless furiously ideas
 - b. *Hiç okumadığı Mehmet kitapları çok Ahmet'in seviyor Turkish
 - This was (and is) a critique of studying meaning alone, or just probabilistically.
- 2 Where is the feel of meaningfulness coming from for senseless expressions?

- Categorical Grammarians have been drawing attention to a very striking asymmetry for **far more than a century**:¹

- Parts of a clause may require different categories,

But one uniquely determines the clausal structure: **The Verb**

If the verb can determine clause's syntactic structure, it can also determine what makes it meaningful.

- Maybe we can put **grammatical categories**, which are grammar's workhorses, to work on BOTH aspects.

¹Husserl (1900); Sapir (1921); Ajdukiewicz (1935); Montague (1973); Steedman (2000); Schmerling (2018); Bozşahin (2025)

- Expressions with sense have intuitive world models (decision models, truth-conditional semantics).
- Expressions with no sense may have counter-intuitive but possible world models (possible world semantics).
- But these do not explain the feel of meaningfulness; they eschew the role of **category choice** in grammaticality:

choosing to refer to another event to the extent of affecting grammaticality.

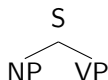
- | | |
|--|----------------|
| (1) a. She played the piano for an hour/*in an hour. | play |
| b. She played the sonata *for an hour/in an hour. | perform |
| c. She played the sonata for a year. | multiple perf. |
| d. She played the piano in a year. | mastery |
| e. She played the piano in an hour. | Genius? |

From distributional labels to abstract categories

Take the following phrase-structure grammar that encodes structure and order, i.e. a context-free grammar:

$$\begin{aligned}
 S &\rightarrow NP \ VP \\
 VP &\rightarrow V_{tv} \ NP \\
 V_{tv} &\rightarrow \text{played}
 \end{aligned}$$

S → NP VP
 VP → V_{tv} NP
 V_{tv} → played



Looking top-down, they appear to be relations.
 Looking bottom-up, we can see that they are actually **FUNCTIONS**.

This is the first step toward understanding the verb's power.

$$VP = S \backslash NP$$

$$NP = S / VP$$

$$V_{tv} = VP / NP = (S \backslash NP) / NP$$

Focusing on the verb, we get for example:

$$played = v_{tv} = (S \backslash NP) / NP : \lambda x \lambda y. play' xy$$

These are structural functions. They determine the tree on the left.

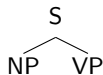
But now we have BOTH syntax and semantics IN A CATEGORY, because they are functionally determined.

S → NP VP

VP → V_{tv} NP VP = S \ NP

V_{tv} → played NP = S / VP

V_{tv} = VP / NP = (S \ NP) / NP



If the subject is ALSO unique in the main clause, why not start with the subject?

NP = S / VP = S / (S \ NP)

Ayşe = S / (S \ NP) : $\lambda p.p \text{ ayşe}'$

Now the predicate is OPAQUE (p). On what basis are we going to choose a category?

This asymmetry is the somewhat neglected discovery of Richard Montague.

- A clause can take many different kinds of constituents with varying categories.
- But one determines what it means to be a clause: The verb
- Since the verb is a predicative element, it can determine the predicate-argument structure (PAS) of the clause.

PAS: Reference Choice for events: Placeholders *then* fill in truth-conditional meanings. Gestaltists might find the idea familiar.

- (2) a. played = $(S \backslash NP) / NP : \lambda x \lambda y . \text{play}' xy$
 b. played = $(S \backslash NP) / PP_{\text{schdl}} / NP_{\text{score}} : \lambda y \lambda x \lambda z . \text{perform}' (iter' x (\text{play}' yz)) z$

- Without that kind of meaning, it is difficult with categories to study grammaticality AND the consequent sense of meaningfulness.

What must syntactic categories do?

- Assemble the PAS as syntactic structure is built.
- Syntax is still autonomous. But now it carries a baggage. The treasure is in the baggage.
- If we don't transmit both structures as we analyze, we would need independent solutions to grammaticality and meaningfulness.
- THAT is not very congenial to modeling. Real data shows why.
- Syntax and semantics can 'stretch' in various ways, **if** conceived independently.

Syntactic decomposition does not necessarily mean semantic decomposition. It is verb-controlled.

- (3) a. Zhāngsān shēng qì le
 Zhangsan generate air asp
 'Zhangsan got angry.'

Chinese; Kao 2024:1

- b. Zhāngsān shēng le hǎodà de qì
 Zhangsan generate asp huge nom air
 'Zhangsan got very angry.' (lit. 'Zhangsan generated huge air.')

- c. Zhāngsān shēng wán qì le
 Zhangsan generate finish air asp
 'Zhangsan stop being angry.' (lit. 'Zhangsan finished generating air.')

Semantic idiomaticity does not mean syntactic inertness. That too is verb-sense-controlled.

(4) a. Wǒ yōu le Zhāngsān yí mò
 I ASP Z one
 'I teased Zhāngsān.'

b. *Wǒ yōu (le) yí mò Zhāngsān

c. *Wǒ yōumò (le) Zhāngsān

To mean 'tease', there has to be an object in between (a–b), light or heavy. Together the verb is not transitive (c).

When we make a category choice, i.e. choice of event reference, we also lay out under what conditions we see grammaticality.

- (5) a. She played the piano for an hour/*in an hour.
 b. She played the sonata *for an hour/in an hour.
 c. She played the sonata for a year.
 d. She played the piano in a year.
 e. She played the piano in an hour.

- (6) a–b. played = $(S \backslash NP) / NP : \lambda x \lambda y. \text{play}' xy$
 c. played =
 $(S \backslash NP) / PP_{\text{schdl}} / NP_{\text{score}} : \lambda y \lambda x \lambda z. \text{perform}'(\text{iter}'x(\text{play}'yz))z$
 d–e. played =
 $(S \backslash NP) / PP_{\text{duration}} / NP_{\text{tool}} : \lambda y \lambda x \lambda z. \text{practice}'(\text{iter}'x(\text{play}'yz))z$

Case (e) is not a whole lot different than searching for meaningfulness for the Chomsky example:

Colorless green ideas sleep furiously.

Roman Jakobson's take: "If someone's hatred never slept, why then, cannot someone's ideas fall into sleep?"

sleep = $S \backslash NP : \lambda x. \text{torpid}' x$

Yue Ren Chao's take:

sleep furiously = $S \backslash NP : \lambda x. \text{sleep-on-idea}' x$

- If syntax-semantics is so dependent on the verb,
and verbs tend to be very choosy about the roles of arguments,
- can we address GENERAL problems of syntax-semantics with this way of thinking?

I suggested in (Bozşahin, 2025) that we can, for example, CASE:

- (7) a. Mary would *to run/*runs/run.
 b. Mary wants *run/*runs/to run.
 c. Mary *run/*to run/runs.

Grammaticality vs. number of readings

- (8) a. Her çocuk araba-ya bin-di. Turkish
 every child-DAT board-PAST
 'Every child got in the car.'
 b. Araba-ya her çocuk bin-di.

How many cars in (a)?

How many cars in (b): Arabaya her çocuk bindi?

Should grammar say something about the number of readings?

- If we want our grammars to take care of grammaticality and the consequent sense of meaningfulness, we can start with the verb in **any** language.
- That's where the choices of event reference and meaning are.
- Leaving all that to implicature, pragmatics or discourse would not explain narrow behavior.
- Sandra (1998) warned us about the two extremes (one lexical entry serves all, all senses served by different entries).
- Without narrow behavior, scientific modeling is hopeless.
- Without an independently replicable path to go from theory of grammar to models of grammar, a 'theory' would not be a natural science theory.

The knowledge of grammar must be representable to be modelable.

- Category choice is an intentional (and intensional) act.
- Only subjects do that; models or computers don't.
- Sapir 1949:17–18 called it *shared psychological spaces*.
- He didn't call it *shared psychological states*.
- We are talking about social semiotics, not psychology.
Accomplishments and achievements are *shared human values*.

- We may not be at the mercy of our psychological or biological make-up for categories. At least Schopenhauer (1819); Ryle (1937) didn't think so.
- To understand the role of choice in **linguistic analysis**, we must worry about how and where the relevant knowledge goes in the grammar.
- Unfortunately, it does not mean we can **express** all that can be imagined.
 - There are GAPS in the linguistic data.
 - That is why linguistics is a natural science.
- It is hard to study common timeframe of language acquisition without such gaps.

- There seems to be **invariants** in the structuring of grammaticality and the consequent sense of meaningfulness (e.g. function composition as math of connectedness).
- These invariants may spell the landscape of the **variants**.
 - Cross-linguistic and intra-linguistic typology.
 - We may discover new ideas to go in to any grammar to explore these aspects.

- Mathematics is the time-honored study of invariants.
- If we do not constrain the space of the variants, we would be in the dark about the sufficiency of supposedly necessary mechanisms.

In short:

There seems to be a limit about the space of possible human linguistic categories. Understanding the limit means understanding ourselves.

- Not quite minimally, all and only the knowledge that affects grammaticality and the consequent sense of meaningfulness (not necessarily sensefulness), enters any grammar.
- That doesn't sound to me like everything under the sun.
- Natural Grammars must be modelable, as a consequence of a theory of grammar.
- A scientific model prepares a theory or an idea for experiments.
- For that we need an explicit nomenclature and modeling vocabulary.
 - Either we make everything under the sun testable this way,
 - Or we narrow the knowledge scope of possible NL grammars so that we can do this.

- Command diversity: possible correspondences of syntactic command and semantic command.
- All possible correspondences (i.e. variants) have been attested. We look at every case in detail.

NB. notes on command diversity

- The invariant's admissibility is necessarily flexible.
- Whether it is sufficiently constrained requires targeting **in the variant** timeframe of **language acquisition**, and limits of **linguistic diversity**.
- Lessons learned in science, starting with Poincaré (1905); Heisenberg (1958):
 - Invariants **admit**. They do not compute. Variants do not admit. They **compute**. Maybe.

- Monadic Maxim (Bozşahin 2025): All linguistic structures are homomorphic to the **structure of analysis**.
- Analysis is based on **connectedness**, mathematically **function composition**, as the only invariant. **NB. notes on analysis**
- The consequent variants bear **categories** representing syntactic and semantic command.

morphology: construction of reference

syntax: transmission of reference

semantics: what is carried in the transmission

- **headed construction**: a **functional category** determines syntax and semantics. These are categories as functions, not contentless labels.
- **non-headed construction**: a **relational category** determines syntax and semantics.
- **Case, agreement, grammatical relations**: **cross-categorical relations** derivable from verbs alone.

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